Thirsty Machines

Visualising the per-prompt environmental cost of generative AI

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Abstract

Public discussion of generative AI's environmental cost trades almost exclusively in aggregates — terawatt-hours per year, billions of litres of water — at scales too large to feel. *Thirsty Machines* starts at the prompt instead, joining five public datasets into a system that traces per-query energy, water, and carbon costs upstream into electricity generation and cooling, and outward across the geographies that absorb them. Eight visual components on a deployed static site cover 66 frontier large language models and 1,479 US data-centre facilities, anchored by an original D3.js animation that lets the reader feel the difference between a 0.27-Wh open-weight query and a 49.93-Wh frontier-reasoning one. A paired white-hat / black-hat static treatment closes the project by showing how the same data supports two radically different conclusions depending on framing.

1 Introduction and question

The environmental cost of generative artificial intelligence is widely reported and poorly understood. Headlines name terawatt-hours and billion-litre figures; readers accept the numbers without being able to picture them. The unit of analysis matters: aggregates compress the choices that produced them, and a person who has just sent a prompt has no way to connect that act to a kilowatt-hour figure published at planetary scale.

This project takes the opposite frame. Its central question is:

> *What does it actually cost the planet when you press send on an AI prompt — and where does that cost land?*

The question is deliberately user-facing. It treats the individual prompt as the unit of analysis and traces its consequences upstream into electricity generation, water cooling, and atmospheric emissions, and outward across geographies. Three sub-questions structure the analysis. *How much does one query actually consume?* — energy in watt-hours, water in millilitres, carbon in grams of CO_2, broken down by model and query type. *Where does the consumption physically occur?* — which cloud regions, which electricity grids, which water basins. *What happens when one query becomes a billion?* — how per-unit costs aggregate across populations and time, and how efficiency gains interact with demand growth.

The intended audience is an informed but non-technical adult who uses generative AI tools regularly, not assumed to know what a kilowatt-hour means in real terms, what data-centre cooling looks like, or what carbon intensity of a grid is; the system defines those terms in place. The audience is, however, assumed to be capable of distinguishing a manipulated chart from an honest one when the manipulation is shown alongside the original — which is the basis of the white-hat / black-hat reflection in Section 6.

2 Data

The system draws on five public datasets, joined into a single analytical surface. All are openly licensed under Creative Commons or the Open Database License, and all were re-downloaded for this project rather than relied on at second hand. Table 1 summarises them.

Table 1: The five primary datasets joined into the project's analytical surface.

Dataset	Source	What we use	Licence
Per-query inference benchmark	Jegham et al., *How Hungry is AI?* (arXiv:2505.09598 v6, Nov 2025)	66 frontier LLMs × 3 query lengths; per-query Wh, mL, g CO_2; PUE, WUE, CIF	CC-BY 4.0
ML systems database	Epoch AI *Notable Models* (epoch.ai/data, weekly)	3,509 systems since 1950; FLOP, parameters, hardware, training power	CC-BY 4.0
Electricity carbon intensity	Ember Yearly + US States (twice-monthly)	228 countries and 50 US states; g CO_2/kWh through 2025	CC-BY 4.0
US data-centre atlas	IM3 Open Source Atlas, PNNL (Feb 2026)	1,479 US facilities; footprint, lat/lon, operator	ODbL
Water-stress indicators	WRI Aqueduct 4.0 (country + provincial)	Baseline water stress on a 0–5 scale by country and US state	CC-BY 4.0

The Jegham benchmark (Jegham et al., 2025) is the spine: it is the only public source that estimates per-query energy, water, and carbon for a wide cohort of frontier large language models in a methodologically uniform way. It was chosen over Luccioni et al.'s earlier "Power Hungry Processing" study (Luccioni et al., 2024) (fewer models, pre-dates the GPT-5 and Claude 4 generations) and over individual provider sustainability disclosures (which never report competitor models). The remaining four sources contextualise it: Epoch supplies training-side temporal context, Ember converts model energy into geographic carbon footprints via per-country and per-state g CO_2/kWh intensity, the IM3 Atlas geocodes US data-centre facilities, and WRI Aqueduct overlays baseline water stress on the same geography.

The data flows through a three-layer pipeline. Phase 1 produces a bronze layer of 11 schema-validated parquet snapshots — faithful to the raw sources, no analytical transformation. Phase 2 applies the documented preprocessing (the AWS multipliers' `skiprows=1` quirk, Aqueduct's -9999 sentinel for Singapore, Ember's filter to `Variable == "CO2 intensity"`, IM3 layer filtering) and produces two integrated silver tables. The first, `models_enriched.parquet` (194×105), holds one row per Jegham model and query length, joined to Epoch training metadata. The second, `regions_geographic.parquet` (37×12), holds one row per AWS or Azure cloud region, joined to Ember carbon intensity, Aqueduct water stress, and IM3 footprint aggregates at state grain. Each layer transition is documented in a single Jupyter notebook, and bronze and silver outputs regenerate reproducibly from those notebooks.

The integration's smoke test is the Virginia / Washington / Arizona triangulation. The same workload routed to Virginia (319 data-centre facilities, 327 g CO_2/kWh grid intensity in 2025, baseline water stress 2.56) carries roughly 2.6× the carbon footprint and meaningfully higher water-supply pressure than the same workload routed to Washington (82 facilities, 124 g CO_2/kWh, water stress 1.20). Arizona — 65 facilities, 283 g CO_2/kWh, water stress 4.39, the highest of the three — is the project's canonical "hot grid in a water-stressed region" case, and motivates the geographic reflection in Section 6. All triangulation numbers reproduce the memo-cited figures within rounding precision.

Five limitations are disclosed throughout the system. *(i)* Jegham estimates are inferred from public API performance, not measured directly: proprietary providers do not publish per-query energy. *(ii)* Geographic attribution is coarse — hyperscalers route inference dynamically, so

the cloud-region map shows *the distribution of regions a provider uses*, not where any specific query went. *(iii)* Hardware efficiency improvements are recent and unevenly applied: 2023- and 2025-vintage figures differ by up to two orders of magnitude. *(iv)* The benchmark covers text-prompt inference for 66 frontier LLMs only — no image, video, or agentic workflows. *(v)* Hong Kong and Singapore return missing values for water stress under Aqueduct's hydrological-basin methodology and are rendered as "no data" tiles.

3 Audience, tasks, and design rationale

From the audience framing the project derives five analytical tasks that the system must support: *Quantify* the per-query cost in human-comprehensible units (watt-hours, millilitres, grams of CO_2); *Compare* that cost across models, providers, query lengths, and hardware tiers; *Locate* where the cost lands geographically by joining model deployment to cloud regions, electricity grids, and water basins; *Scale* the per-query unit to aggregate consumption at one million, one billion, and one hundred billion queries with everyday equivalences; and *Reflect* on the analytical tension between rapidly improving per-query efficiency and rapidly growing aggregate demand. The eight visual components, enumerated in Section 4, are each scoped to one or more of these tasks. No component is decorative.

Organisation identity is encoded with a single colour palette across the entire system. The project uses Wong's eight-colour palette (Wong, 2011), originally designed for accessibility under colour vision deficiency, mapped one-to-one to the seven providers in the dataset (Meta, OpenAI, Anthropic, DeepSeek, Mistral AI, Google, xAI). The same hex codes drive the Tableau workbooks, the D3 component, and the Python statics, so a reader moving across the system never has to rebuild a colour-to-provider mapping. An initial mapping that collided Google and xAI on a shared grey was caught during white-hat construction and corrected; the palette is now consistent across all three rendering ecosystems.

The project instructions impose a strict "maximum one line or bar chart" rule across the visual system. The team's reading is the strict one: at most one chart total, of either plain-line or plain-bar form. Component 4 (training compute trajectory) takes that slot. Every other component uses a non-bar, non-line form: a ranked dot plot, small-multiple traces, pictograms with everyday equivalences, a Sankey, a global cloud-region map, an animated pipeline, and inventive paired statics. The constraint shaped most of the design decisions and is worth surfacing because the exploratory notebook prototypes used convenience matplotlib forms (including bars) for feasibility-checking that the production system never adopts.

4 System description

The system is deployed as a static site at `https://thirsty-machines.vercel.app/`, with continuous deployment from the project's public GitHub repository.[1] Eight visual components carry the analysis, summarised in Table 2. Five are interactive and linked, above the brief's four-interactive minimum for a two-person group; one is the original D3.js animation that anchors the project; one is the paired white-hat / black-hat static; and one is a Sankey system map. The JavaScript animation simultaneously satisfies three brief requirements — it is the project's original JavaScript component, its creative visualisation, and its explanatory view for non-experts.

The components cohere through shared structural patterns rather than through a single dashboard. The Wong palette identifies organisations consistently across Components 1, 2, 6, and 7. Component 1's region callout, Component 3's geographic map, and Component 8's Sankey all surface the same Virginia routing motif — Azure East US, 327 g CO_2/kWh, water stress 2.56 —

[1] Source: `https://github.com/IIxoskeletonII/Thirsty-Machines`. The README lists the six Tableau Public workbook URLs.

Table 2: The eight visual components of the deployed system. Interactive components are linked within their dashboards via model and query-length filters.

#	Component	Form	Tool	Role
1	Prompt-pipeline animation	Interactive	D3.js + JS	Centerpiece; original JS component, creative visualisation, explanatory view for non-experts
2	Model energy ranking	Interactive	Tableau	Ranked dot plot with small multiples by query length; 184× long-tier spread
3	Cloud region map	Interactive	Tableau	37 AWS + Azure regions; grid carbon intensity and water-stress overlays
4	Training compute trajectory	Interactive	Tableau	The one allowed line chart; n=15 frontier-disclosed FLOP
5	Aggregate scaling	Interactive	Tableau	Pictogram; per-query cost projected to 1M, 1B, 100B queries with everyday equivalences
6	Efficiency vs. query length	Interactive	Tableau	Per-model small-multiple traces across the three query lengths
7	White-hat / black-hat pair	Static	Python	Methodological-literacy reflection; same data, two narratives
8	System map (Sankey)	Interactive	Tableau	Query → model → region → grid → emissions flow

at three different units of analysis: per-query, per-region, and per-flow. Filtering is honoured at the dashboard level: Components 2, 3, 5, and 6 each carry model and query-length filters that link within the dashboard. Cross-dashboard linkage is left to the narrative structure of the site, which walks the reader through the components in task order rather than tab order: *Quantify* (C1), *Compare* (C2, C6), *Locate* (C3, C8), *Scale* (C5), *Reflect* (C7), with Component 4 supplying training-side historical context.

5 Rejected alternatives

Two design alternatives were considered seriously before settling on the current system, and one chart design was reframed mid-project after exploratory verification.

The first rejected alternative was **Power BI as the BI tool**. The Jegham per-query benchmark is itself published as a Power BI dashboard, which made Power BI a natural starting choice. It was rejected because Microsoft's free public-publishing tier is being deprecated, embedding requires a paid Pro license the team did not have, and the team had stronger prior exposure to Tableau. Reusing the source dataset's BI tool would also have created presentation overlap rather than analytical contribution.

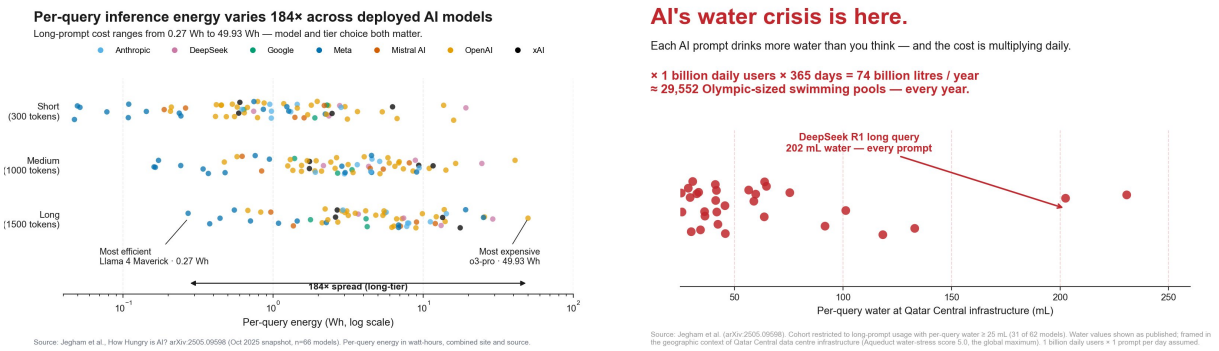
The second rejected alternative was **a pure-Python interactive dashboard via Plotly Dash or Streamlit**. This would have unified the stack and removed the Tableau dependency. It was rejected because the brief counts Tableau and Power BI as ecosystem options toward the three-ecosystem minimum, while Python and JavaScript alone count as two. Choosing Python-only would have forced heavier reliance on the JavaScript component to satisfy the ecosystem requirement, increasing risk on the project's most novel piece.

Within the system itself, the original **Component 2** design — a scatter on Epoch FLOP or Parameters — was reframed during Phase 3 after coverage verification. FLOP populated for only 15 of 66 unique Jegham models, Parameters for only 12, with three of the six prompt-pipeline showcase models absent from both axes. An Epoch-axis scatter would have excluded most of the systems the dashboard otherwise spotlights. The reframe to a ranked Jegham dot plot at 100% per-query coverage preserved the analytical intent — where models sit on the energy spectrum — without the coverage problem.

6 Findings and limitations

Four findings emerge. **(1)** Per-query inference energy varies by approximately 184× across deployed models at long-tier — from Llama 4 Maverick at 0.27 Wh to o3-pro at 49.93 Wh per long query. Ranking is stable at the extremes (frontier reasoning pinned expensive, lightweight pinned cheap) but reshuffles in the middle as queries lengthen, with mid-tier models shifting twenty-plus positions between short and long prompts (Spearman ρ short $\leftrightarrow$ long 0.893). **(2)** Hosting choice matters as much as model choice: DeepSeek R1 served by the vendor versus via Azure shows roughly 4× energy and 7× carbon variance on identical workloads. **(3)** Geographic distribution is uneven — the Virginia–Washington 2.6× carbon ratio of Section 2 extends to Arizona, where US-high water stress (4.39) combines with a mid-range grid intensity (283 g CO_2/kWh). **(4)** Per-query efficiency is improving substantially while aggregate demand grows faster still; both statements are simultaneously true, and their reconciliation is a Jevons-paradox argument the paired statics make structurally explicit.

White-hat / black-hat reflection. Figure 1 renders the same per-query data under two framings. Panel (a) is honest: log-scale, full sample of 66 models, neutral palette, the 184× long-tier spread annotated against the two anchor models. Panel (b) applies six legitimate-looking manipulations to push the opposite conclusion: *sample restriction* (only the 31 of 62 long-tier models with per-query water $\geq$ 25 mL — the heavy half, the efficient half excluded by construction); *axis truncation* (x-axis starts at 25 mL); *geographic recontextualisation* (Jegham's globally-distributed values framed as Qatar Central, Aqueduct water-stress 5.0, through the x-axis label only — every value remains as published); *aggregate framing* (per-query water projected to one billion daily users × 365 days, producing 74 billion litres per year, expressed as 29,552 Olympic pools); *palette and typography* (deep red primary, bold heavy weights, newsroom register); and *source de-emphasis* (attribution at 7 pt grey). The Qatar manipulation was originally a WUE rescale, rejected because Qatar Central's site WUE of 1.29 is *below* the global mean of 4.6 and would have reduced visible water values; the framing-only replacement is a richer technique — every value remains as published, and the dishonesty is purely structural.



(a) White-hat (b) Black-hat

Figure 1: Same per-query data under two framings: **(a)** the white-hat renders the distribution honestly, log-scale and full-sample, with the 184× long-tier spread annotated; **(b)** the black-hat applies six legitimate-looking manipulations (sample restriction, axis truncation, geographic recontextualisation, aggregate framing, palette and typography, source de-emphasis) to push the opposite conclusion from the same underlying data.

Three additional limitations surfaced during integration: Anthropic-self-hosted, Google, xAI, and DeepSeek vendor-hosted rows lack per-region infrastructure multipliers; Epoch's `Training power draw (W)` is empty across all 134 matched rows (Components 2 and 4 use training compute and parameters instead); Hong Kong and Singapore return "no data" tiles for water stress under Aqueduct's basin methodology. Future work: GCP regions, sub-state grid zones, and image,

video, and agentic-workflow benchmarks.

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AI Use Appendix

Tools used

Two AI tools supported the project, in distinct phases.

Gemini was used at the pre-implementation stage to examine the candidate datasets named in the project brief and to verify project feasibility before any code was written. The output of that examination drove the Section 2 dataset selection and surfaced the brief edits required before implementation began.

Claude Sonnet 4.7 was used during implementation for webpage construction (the HTML, CSS, and JavaScript scaffolding of the deployed static site), code quality checks (notebook cleanups and refactoring of redundant patterns), removal of dead code and verbose comments, prose polishing for the report and the README, and information retrieval (locating specific facts and figures from the accumulated project documentation when drafting report prose).

Accountability

All commits were authored manually by Eliya Allam under the GitHub no-reply email. Both team members reviewed all AI-suggested code and prose in full before adoption; stylistic markers typical of unrevised AI output (decorative comment headers, emoji, verbose docstrings on trivial functions, over-defensive error handling) were stripped during review. No AI-tool references appear anywhere in the committed code, comments, or commit messages. The team — Eliya Allam and Mattia Cervelli — is jointly responsible for every sentence, figure, code block, visual decision, and conclusion in the submission, and both members are prepared to defend any element of the work.

Representative prompts

(1) Gemini — dataset feasibility verification (pre-flight). Asked Gemini to examine each candidate primary dataset named in the draft brief and produce, per dataset, a verdict against four criteria: *is it publicly downloadable; are the licence terms compatible with public academic use; are the columns the brief assumes actually documented in the schema; do the join keys to the other datasets resolve cleanly?* Output: a structured per-dataset feasibility memo with verdicts of *Confirmed*, *Conditional*, or *Reduced scope*, plus the specific brief edits required. *Accepted:* the dataset-by-dataset structure, the LBNL reclassification (see worked example below), and the documentation of the IM3 facility-footprint-versus-capacity caveat. *Rejected:* an assertion that the Aqueduct download required preprocessing at ingestion — the column documentation confirmed the file is usable as published.

(2) Claude Sonnet 4.7 — site narrative composition. Asked Claude to compose the deployed site's section structure, the connective prose between embedded components, and the inline-defined terminology for non-expert readers (frontier large language model, token, inference, g CO_2/kWh, water-stress score, FLOP). The prompt included the eight-component specification, the four findings, the dataset list, and the audience framing. *Accepted:* the nine-section narrative structure, the inline term definitions on first appearance, and the four-findings ordered list. *Rejected:* an initial draft that overused "we" across all sections — the team replaced it with a third-person analytical register, reserving "we" only for the explicit methodology disclosure paragraph.

(3) Claude Sonnet 4.7 — code polish and redundant-code removal. Pasted the assembled notebook and component source files to Claude and asked it to flag: *(i)* redundant helper functions duplicated across notebooks; *(ii)* variable assignments that were never read; *(iii)* verbose docstrings on functions whose name and signature already explained the behaviour;

(iv) comments that explained *what* the code did rather than *why*. Output: a structured set of suggested changes with rationale per change. *Accepted:* the docstring trimming and the removal of unread assignments. *Rejected:* a proposed refactor that would have combined unrelated logic into a single multi-purpose function — the readability cost outweighed the line-count saving.

Worked example: weak or wrong output and the correction applied

The Gemini feasibility pass examined the brief's classification of the LBNL 2024 *Data Center Energy Usage Report* as a primary structured dataset — the same tier as Jegham, Epoch, Ember, IM3, and Aqueduct. On inspection of LBNL's publication channels, Gemini surfaced that the report is distributed only as a PDF: no supplementary CSV, no public API endpoint, no GitHub data repository. The "primary dataset" framing was therefore wrong, because LBNL cannot be ingested into the same bronze–silver parquet pipeline as the structured sources. The team reclassified LBNL from primary to reference: specific figures from the LBNL report are cited inline in the project's documentation and a single small reference table (`data/reference/lbnl_us_dc_energy_2014_2028.csv`) was manually transcribed for the figures the project actually consumes. The pattern derived from this catch: *"available as a public document"* and *"structured-data-pipeline ingestable"* are not the same property, and a feasibility check must distinguish them explicitly before any ingestion code is written.